

## 2.4 Global Governance of the Earth's Oceans

Geography — Eduqas A-Level (A110QS) — Component 2: Global Governance

### 1. The Ocean as a Global Commons

The ocean is both a physical system and a political problem. It covers roughly **71% of the Earth's surface**, contains about **97% of the planet's water**, and — because the great ocean basins (Pacific, Atlantic, Indian, Arctic, Southern) are interconnected through straits, channels and meridional circulation — is best conceptualised as a single *world ocean*. That physical unity collides with political reality: the world is divided into 193 sovereign states plus disputed territories, and the ocean simultaneously falls under national jurisdiction (within 200 nautical miles of coastlines), shared jurisdiction (transit zones, regional fisheries) and no jurisdiction at all (the high seas, which make up nearly **64% of the ocean area** and **43% of the Earth's surface**). Governing a resource that is biologically connected, politically fragmented and largely invisible from land is the defining challenge of the twenty-first century.

The ocean provides four categories of value that are central to any governance discussion. First, it is a **natural resource base** — fisheries supply ~17% of the world's animal protein; the seabed contains oil, gas, sands, polymetallic nodules and rare-earth-bearing crusts. Second, it is a **transport medium**: ~80% of global merchandise trade by volume moves by sea, via ~50,000 large commercial vessels. Third, it is a **climate regulator** — absorbing ~30% of anthropogenic CO<sub>2</sub> emissions and more than 90% of the additional heat trapped in the Earth system since 1970. Fourth, it provides **ecosystem and cultural services** — coastal protection, tourism, recreation, spiritual and indigenous value, biodiversity stewardship.

**Global commons:** Resource domains that lie beyond the exclusive jurisdiction of any single state and are, in principle, open to use by all — the high seas, the deep seabed ("the Area"), the atmosphere, Antarctica, and outer space. Their governance requires multilateral cooperation because no single state can either protect them or be excluded from them.

Because no one owns the high seas or the deep seabed, they are vulnerable to the classic *tragedy of the commons* (Hardin, 1968): individual actors, pursuing their own short-term interest in fishing, dumping or extraction, collectively degrade a shared resource. The 20th-century record — collapsing fish stocks, garbage patches, dead zones, the 1982 negotiation of a “constitution for the oceans” — is largely a story of attempts to avoid precisely this outcome.

## 2. The Ocean as a Physical System

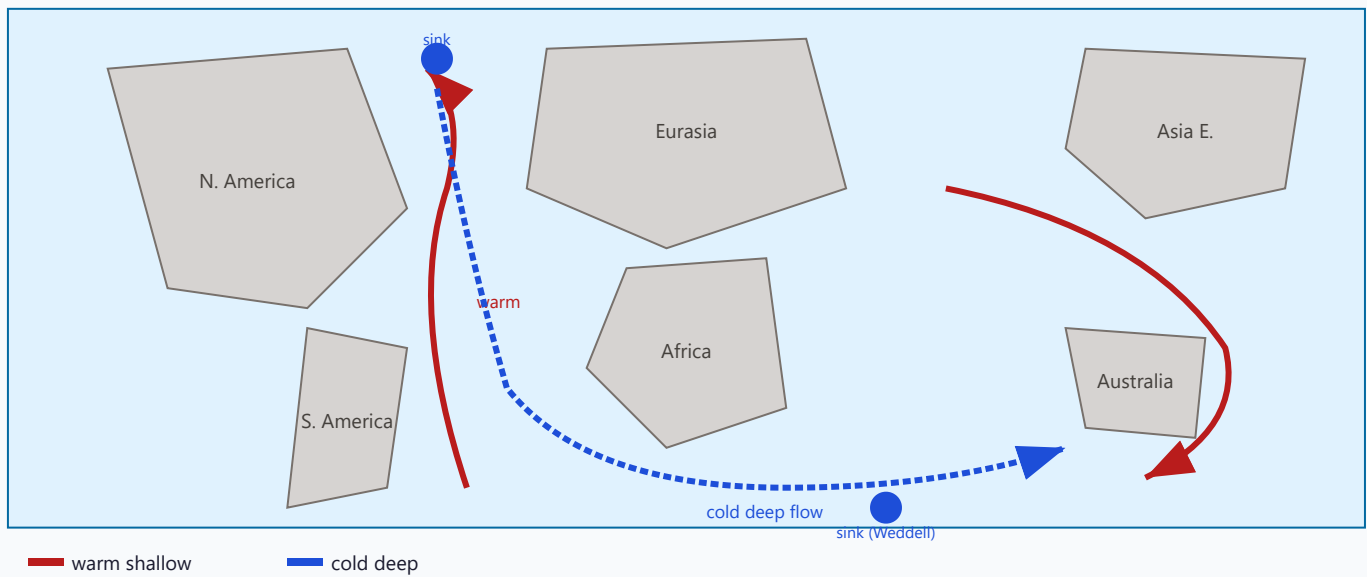
Global governance cannot be understood without the physical geography of the ocean. Four connected processes matter especially.

### Thermohaline circulation (the “global conveyor”)

Differences in water *temperature* and *salinity* drive a global-scale, three-dimensional circulation. Cold, salty water sinks in the North Atlantic (south of Greenland) and the Weddell Sea off Antarctica; this dense bottom water flows southward and then eastward, rising slowly in the Indian and Pacific Oceans before returning as warmer surface water. A full circuit takes ~1,000–1,500 years. The North Atlantic limb, the Atlantic Meridional Overturning Circulation (AMOC), carries heat northward and is the reason north-west Europe is ~5 °C warmer than it would otherwise be. Freshwater input from Greenland ice melt is weakening the AMOC — the IPCC’s Sixth Assessment Report (AR6) (2021) judged an abrupt collapse before 2100 unlikely (medium confidence), though some later studies suggest a higher risk, and the consequences would be severe.

**Thermohaline circulation:** The slow, global-scale overturning of the oceans driven by density differences arising from temperature (*thermo-*) and salinity (*-haline*). It redistributes heat from the tropics to high latitudes and is a key regulator of climate on multi-decadal to millennial timescales.

### Thermohaline Circulation — Global Conveyor Belt



## Surface gyres and upwelling

Wind-driven surface currents organise each ocean basin into large **gyres** — the North Pacific, South Pacific, North Atlantic, South Atlantic and Indian Ocean gyres. These are clockwise in the northern hemisphere and counter-clockwise in the southern hemisphere, a consequence of the Coriolis effect. Each gyre has a quiet, slow-moving centre where floating debris accumulates; the *North Pacific Subtropical Gyre* is the home of the “Great Pacific Garbage Patch”. Along the eastern margins of the major basins — off Peru–Chile, Namibia–Angola, Mauritania–Morocco and California — cold deep water rises in *upwelling zones*. These cover only ~1% of the ocean but provide ~25% of global capture fisheries.

## Marine biomes

Four biomes recur in exam questions. (1) **Coral reefs** (~0.1% of the ocean but home to ~25% of marine species) are warm-water ecosystems built by symbiosis between corals and photosynthetic algae (zooxanthellae); they are extraordinarily vulnerable to temperature stress. (2) **Mangroves** line tropical and subtropical coasts, provide nurseries for ~80% of commercial fish species, and store ~3–5x more carbon per hectare than tropical rainforest (“blue carbon”). (3) The **deep sea** (below 200 m) is the largest biome on Earth by volume, with hydrothermal vents hosting chemosynthetic communities independent of sunlight. (4) **Polar ocean** ecosystems are short, high-biomass food chains anchored on krill and adapted to sea ice.

## Ocean as carbon sink and heat reservoir

The ocean has absorbed roughly **30% of all anthropogenic CO<sub>2</sub>** released since 1750 (~160 gigatonnes) and more than **90% of the excess heat** accumulated in the climate system since 1970. Without this buffer, surface warming would be far greater than observed. The costs are ocean warming, *acidification* (surface pH has fallen ~0.1 units, equivalent to ~30% more acidic), deoxygenation (global mean dissolved oxygen has fallen ~2% since 1960), and sea-level rise (~3.7 mm per year since 2006, driven roughly equally by thermal expansion and ice melt).

### 3. UNCLOS 1982 — the Constitution for the Oceans

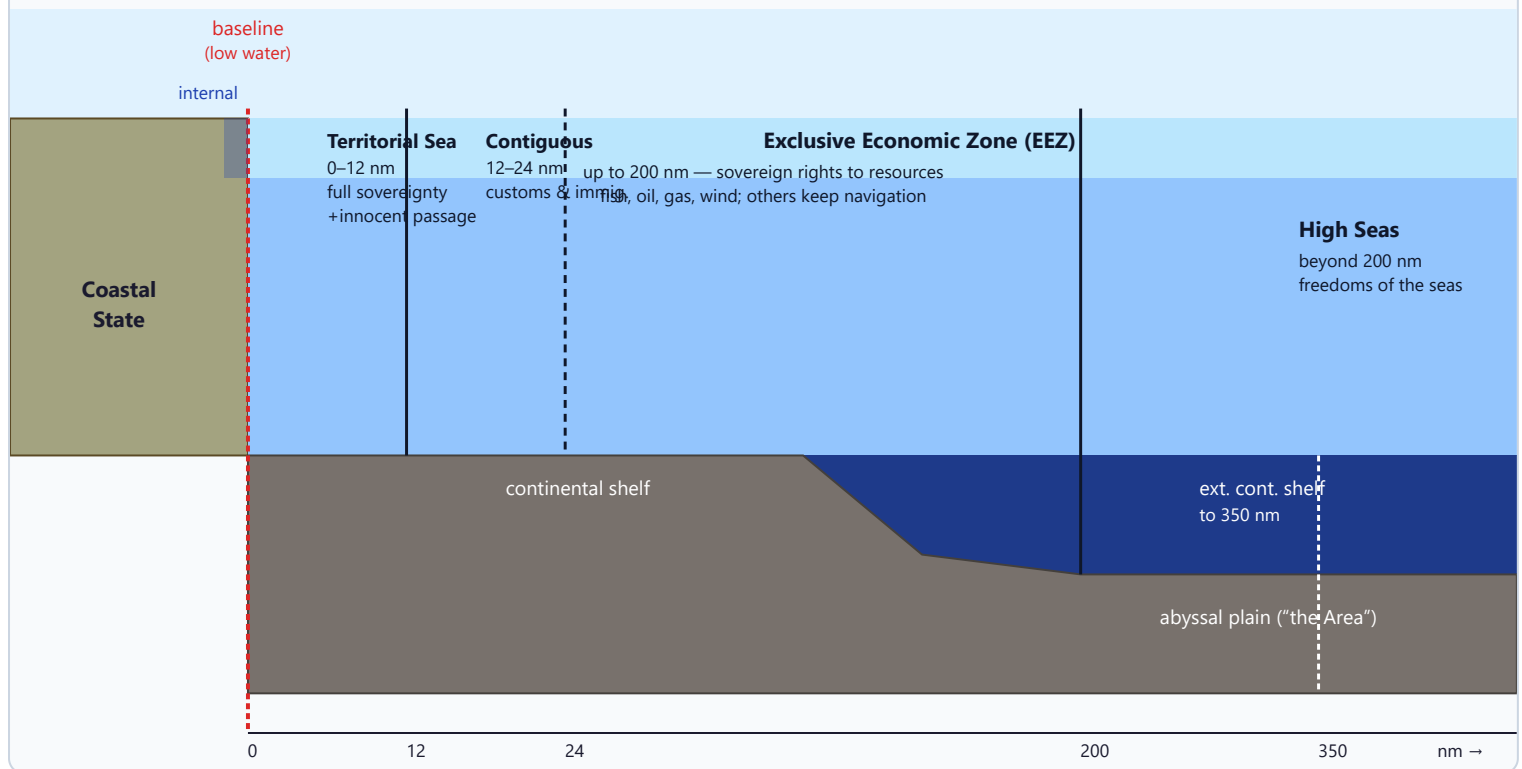
The **United Nations Convention on the Law of the Sea**, adopted at Montego Bay in 1982 and entering into force on 16 November 1994 after the 60th ratification, is the single most important legal instrument for ocean governance. UN Secretary-General Javier Pérez de Cuéllar called it “possibly the most significant legal instrument of this century”, and Tommy Koh, president of the negotiating conference, called it “a constitution for the oceans”. **169 parties** (168 states plus the European Union) had ratified by 2024. The United States has neither signed nor ratified the Convention (it signed only the 1994 Implementation Agreement), relying instead on customary international law.

**UNCLOS (United Nations Convention on the Law of the Sea, 1982):** The comprehensive multilateral treaty setting out the legal framework for all uses of the oceans and their resources. It defines maritime zones (territorial sea, contiguous zone, EEZ, continental shelf, high seas, the Area), the rights and duties of coastal and flag states, dispute-settlement procedures, and the institutional architecture (ISA, ITLOS, CLCS) that administers them.

UNCLOS replaced a patchwork of 17th-century Grotian “freedom of the seas” doctrine and four 1958 Geneva conventions. Its central achievement was to define nested maritime zones measured outward from a coastal state’s *baseline* (usually the low-water line), with decreasing sovereignty as distance from shore increases.

#### Maritime zones — cross-section

### UNCLOS Maritime Zones (cross-section, not to scale)



## The zones in detail

- **Baselines:** Normally the low-water line along the coast; archipelagic states (Indonesia, Philippines) may draw "straight baselines" around the outermost islands under Part IV.
- **Internal waters:** Bays, rivers, harbours and waters landward of the baseline. Full state sovereignty; no right of innocent passage for foreign vessels.
- **Territorial sea (0–12 nm):** Full sovereignty over water, airspace, seabed and subsoil. Foreign ships enjoy *innocent passage* (continuous and expeditious, not prejudicial to peace or security).
- **Contiguous zone (12–24 nm):** Coastal state may enforce customs, fiscal, immigration and sanitary laws; not a zone of full sovereignty.
- **Exclusive Economic Zone (EEZ, 0–200 nm):** Sovereign rights over exploration, exploitation, conservation and management of natural resources (living and non-living). Other states retain freedoms of navigation, overflight, cable-laying.
- **Continental shelf:** Sovereign rights over seabed and subsoil resources out to 200 nm automatically, and beyond (to a maximum of 350 nm or 100 nm beyond the 2,500 m isobath) where geologically justified — subject to recommendation by the *Commission on the Limits of the Continental Shelf* (CLCS).

- **High seas:** All parts of the sea not within an EEZ, territorial sea or internal waters. Six freedoms: navigation, overflight, laying of submarine cables/pipelines, construction of artificial islands, fishing, marine scientific research.
- **The Area:** The seabed and ocean floor beyond the limits of national jurisdiction. Declared the “common heritage of mankind” (Art. 136); mineral resources administered by the International Seabed Authority on behalf of humanity as a whole.

**Exclusive Economic Zone (EEZ):** A maritime belt extending up to 200 nautical miles (~370 km) from a coastal state’s baselines, within which the state has sovereign rights to explore, exploit, conserve and manage natural resources (fisheries, oil, gas, wind, wave, seabed minerals) but does not have full territorial sovereignty; other states retain rights of navigation, overflight and cable-laying.

**Territorial sea:** The coastal belt, up to 12 nautical miles from the baseline, over which a state exercises full sovereignty subject only to the right of *innocent passage* for foreign merchant and warships not engaged in hostile, fishing or polluting activity.

**High seas:** Ocean areas beyond the 200 nm EEZ of any state — about 64% of the ocean and 43% of the Earth’s surface. Governed by the principle of freedom of the seas (navigation, fishing, overflight, cable-laying, scientific research, construction) subject to treaty obligations.

| Zone            | Distance             | Coastal state rights                            | Other states                   |
|-----------------|----------------------|-------------------------------------------------|--------------------------------|
| Internal waters | Landward of baseline | Full sovereignty                                | No automatic access            |
| Territorial sea | 0–12 nm              | Full sovereignty                                | Innocent passage               |
| Contiguous zone | 12–24 nm             | Enforcement (customs, immig., sanitary, fiscal) | Navigation freedoms            |
| EEZ             | 0–200 nm             | Sovereign rights over resources                 | Navigation, overflight, cables |

|                   |                                     |                                      |                                     |
|-------------------|-------------------------------------|--------------------------------------|-------------------------------------|
| Continental shelf | To 200 nm (or 350 nm extended)      | Sovereign rights to seabed resources | Navigation above shelf unaffected   |
| High seas         | Beyond 200 nm                       | None exclusive                       | Six freedoms of the seas            |
| The Area          | Seabed beyond national jurisdiction | None                                 | Common heritage; ISA administration |

**Exam tip — cite UNCLOS zones correctly:** Examiners reward precise numbers. “Territorial sea 12 nm, contiguous 24 nm, EEZ 200 nm, continental shelf to 350 nm” — memorise the four figures and the principle that sovereignty *decreases* with distance. A diagram sketch of the zones often earns a mark even in extended-response questions.

## UNCLOS institutions

Three bodies administer the Convention:

- **International Seabed Authority (ISA)** — based in Kingston, Jamaica; regulates mineral exploration and eventually exploitation of the Area on behalf of humanity.
- **International Tribunal for the Law of the Sea (ITLOS)** — based in Hamburg; judicial body for UNCLOS disputes (e.g. Bangladesh/Myanmar maritime delimitation, 2012).
- **Commission on the Limits of the Continental Shelf (CLCS)** — advisory scientific body reviewing submissions for extended continental shelves (notably Russia’s and Canada’s Arctic claims).

## 4. The Institutional Landscape

Beyond UNCLOS, a dense network of global and regional bodies regulates specific ocean uses. Exam answers that name two or three of these correctly and link them to particular functions stand out.

| Body                                      | Founded          | Role                                            | Key instruments                         |
|-------------------------------------------|------------------|-------------------------------------------------|-----------------------------------------|
| IMO (International Maritime Organization) | 1948 (UN agency) | Shipping safety, security, pollution prevention | SOLAS, MARPOL, Ballast Water Convention |

|                                                         |         |                                                |                                                                                    |
|---------------------------------------------------------|---------|------------------------------------------------|------------------------------------------------------------------------------------|
| ISA (International Seabed Authority)                    | 1994    | Deep-seabed mining in the Area                 | Exploration contracts, draft Mining Code                                           |
| ITLOS                                                   | 1996    | UNCLOS dispute settlement                      | ~35 judgments (2024)                                                               |
| FAO (Food and Agriculture Organization)                 | 1945    | Fisheries statistics & sustainable aquaculture | Code of Conduct for Responsible Fisheries 1995; Port State Measures Agreement 2016 |
| RFMOs (~17 regional fisheries management organizations) | various | Set catch limits for shared stocks             | ICCAT (tuna Atlantic), CCAMLR (Antarctic), NEAFC (NE Atlantic)                     |
| UN-Oceans                                               | 2003    | Inter-agency coordination within UN system     | Communiqués; SDG 14 reporting                                                      |
| UNEP Regional Seas Programme                            | 1974    | Regional cooperation on pollution & habitats   | 18 Regional Seas conventions (Barcelona, OSPAR, Nairobi)                           |
| CBD (Convention on Biological Diversity)                | 1992    | Marine biodiversity, EBSAs                     | Aichi Targets (2010–20); Kunming-Montreal GBF 2022 (30x30)                         |

**MARPOL (International Convention for the Prevention of Pollution from Ships, 1973/78):** The principal IMO instrument preventing operational and accidental pollution from ships. Six annexes cover oil, noxious liquids, packaged harmful substances, sewage, garbage, and air pollution (including sulphur caps and the 2020 global 0.5% sulphur limit on marine fuel).

## 5. Fishing and Fisheries Governance

Capture fisheries yield around **90 million tonnes** of wild-caught fish annually; aquaculture adds another **~94 million tonnes** (2022), overtaking wild capture in total aquatic animal production for the first time; farmed fish have supplied most of the fish eaten by people since 2014. About **600 million people** depend

on fisheries for their livelihoods. The *FAO State of World Fisheries and Aquaculture 2024* reports that **~37.7% of assessed marine fish stocks are fished at biologically unsustainable levels** (up from 10% in 1974), with the remainder at maximum sustainable yield (~57%) or underfished (~6%).

## Pressures on fisheries

- **Overcapacity:** The global fishing fleet (~4.1 million vessels, 2.9 million motorised) is estimated at 2–3× the size needed to catch the maximum sustainable yield.
- **IUU fishing:** Illegal, Unreported and Unregulated fishing is valued at **\$10–23 billion** per year (11–26 million tonnes). West African waters are particularly affected.
- **Bottom trawling:** Heavy nets dragged across the seabed destroy cold-water coral gardens and seamount habitats; ~5 million km<sup>2</sup> of seabed is trawled annually.
- **Bycatch:** ~40% of global catch is discarded bycatch (40 million tonnes). Gillnets kill ~300,000 cetaceans yearly.
- **Subsidies:** Global fishing subsidies total ~\$35 billion per year, of which ~\$22 billion is “harmful” (fuel subsidies that enable overfishing). The WTO Agreement on Fisheries Subsidies (2022) is a first-step discipline; it entered into force in September 2025.

**IUU fishing (Illegal, Unreported and Unregulated):** Fishing activity conducted in violation of national or international rules (illegal), outside reporting requirements (unreported), or in areas lacking applicable conservation rules (unregulated). Estimated at \$10–23 billion per year globally and a major threat to SDG 14.4.

## The EU Common Fisheries Policy and Brexit

The **Common Fisheries Policy (CFP)** pooled EU member-state waters into a single quota system based on “relative stability”, with Total Allowable Catches (TACs) set annually by the Council. Reformed in 2013 to introduce a Landing Obligation (discard ban) and a commitment to Maximum Sustainable Yield (MSY). The UK’s departure from the EU (2020) removed the UK from the CFP; the 2020 EU–UK Trade and Cooperation Agreement preserved *reciprocal access* for a transition period to June 2026; in 2025 the UK and EU agreed to extend full reciprocal access until 2038. Politically controversial, especially in Scotland and south-west England; economically modest (UK catch value ~£1bn/yr).

**Exam tip — balance commercial and environmental arguments:** Eduqas governance questions often ask you to *evaluate*. Show the trade-off: fishing is a 600-million-person livelihood and 17% of

animal protein globally, but 38% of stocks are overfished. A strong answer quantifies both sides rather than arguing one way.

## 6. Pollution of the Oceans

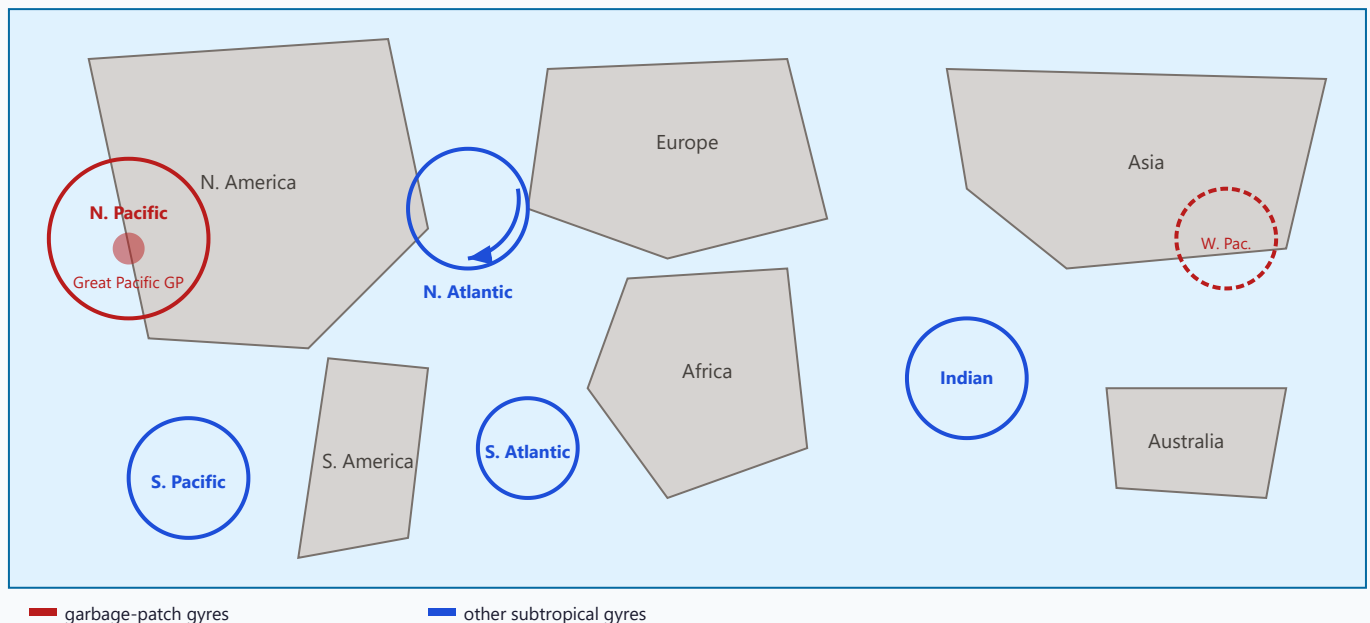
Seven pollution streams recur in exam questions. They differ in source, persistence and the governance tools used to tackle them.

| Pollution type                  | Main sources                                                   | Scale                                              | Governance tools                                                  |
|---------------------------------|----------------------------------------------------------------|----------------------------------------------------|-------------------------------------------------------------------|
| Plastic                         | Land-based (~80%); fishing gear (~10%); shipping               | ~11 Mt/yr entering ocean; 150 Mt accumulated       | UNEP Plastics Treaty (under negotiation); EU Single-Use Directive |
| Oil                             | Tanker accidents; rig blowouts; operational discharges; runoff | ~0.7 Mt/yr (falling)                               | MARPOL Annex I; OPRC 1990; IOPC Funds                             |
| Nutrients (N, P)                | Agricultural runoff; sewage                                    | ~500 dead zones globally, ~245,000 km <sup>2</sup> | Regional agreements (OSPAR, Baltic HELCOM); Nitrates Directive    |
| Chemicals / POPs                | Industrial discharge; airborne deposition                      | PCBs, heavy metals, pharmaceuticals persistent     | Stockholm Convention 2001; Minamata Convention 2013               |
| Noise                           | Shipping; sonar; seismic surveys                               | Surface noise doubled every decade                 | IMO Guidelines 2014; regional CMS                                 |
| Radioactive                     | Nuclear tests (historic); Fukushima 2011                       | Low globally; regional hotspots                    | London Convention 1972 (dumping ban); IAEA oversight              |
| CO <sub>2</sub> → acidification | Fossil-fuel emissions                                          | pH -0.1 since 1850 (~30% more acidic)              | UNFCCC / Paris Agreement 2015                                     |

## Plastic pollution

An estimated **11 million tonnes of plastic enter the ocean every year**, a figure projected to nearly triple by 2040 under business-as-usual. 80% originates from land-based sources — mismanaged municipal waste and rivers draining dense coastal populations. The **Great Pacific Garbage Patch** in the North Pacific Subtropical Gyre covers an estimated **1.6 million km<sup>2</sup>** (roughly 3× the area of France) containing ~1.8 trillion plastic pieces. The patch is predominantly fishing gear (~46% by mass) and microplastics. Microplastics (<5 mm) have been found in deep-sea trenches, Arctic ice, human blood and placenta. A legally binding **Global Plastics Treaty** is under negotiation at UNEP (INC-5 in Busan, November 2024), with a target conclusion date initially set for 2024; neither INC-5 (Busan, November 2024) nor a resumed session in Geneva (August 2025) reached agreement.

Major Ocean Gyres and Garbage Patches



## Oil pollution

Annual oil input to the ocean has fallen from ~3 Mt in the 1970s to ~0.7 Mt by the 2020s, thanks mainly to MARPOL Annex I (double-hulled tankers, segregated ballast, port reception facilities). Accidental spills, though smaller than operational discharges in total, generate the highest-profile governance responses.

### Case study: Deepwater Horizon, Gulf of Mexico (2010)

On 20 April 2010 the *Deepwater Horizon*, a mobile offshore drilling unit operated by Transocean and

leased by BP, suffered a blowout at the Macondo well (~1,500 m water depth, ~66 km off Louisiana). Eleven crew were killed and the rig sank on 22 April. Oil flowed uncontrolled for **87 days** until the well was capped on 15 July and sealed on 19 September. US government estimates put the release at **~4.9 million barrels (~780,000 m<sup>3</sup>)** — the largest accidental marine oil spill in history.

*Environmental impact:* ~2,100 km of coastline in Louisiana, Mississippi, Alabama, Florida and Texas was oiled; hundreds of thousands of birds and many thousands of sea turtles and cetaceans died; deep-sea coral communities and commercial fisheries (oysters, shrimp, menhaden) were severely damaged. The dispersant Corexit (~7 million litres) dispersed surface oil into the water column, with disputed long-term effects.

*Economic and legal:* BP paid ~\$65 billion in clean-up, fines, compensation and settlements, the largest corporate environmental penalty in US history. The 2015 Consent Decree with the US federal government and five Gulf states totalled \$20.8 billion alone. A \$4 billion criminal settlement included the first felony manslaughter convictions for a corporate oil spill.

*Governance response:* The US Bureau of Ocean Energy Management (BOEM) and Bureau of Safety and Environmental Enforcement (BSEE) were created in 2011 to separate leasing from safety regulation. New rules on blowout preventers (Well Control Rule 2016), drilling safety and containment capabilities followed. Internationally, the OPRC Convention gained fresh attention, though deep-water drilling standards remain flag-state-dependent.

## Eutrophication and dead zones

Excess nitrogen and phosphorus from fertiliser runoff and sewage stimulate phytoplankton blooms; when they die and decompose they consume dissolved oxygen, producing *hypoxic* zones (<2 mg O<sub>2</sub>/l) where most fauna cannot survive. There are **~500 identified dead zones** totalling ~245,000 km<sup>2</sup>, including the Gulf of Mexico (Mississippi plume, up to 22,700 km<sup>2</sup>), the Baltic Sea (~70,000 km<sup>2</sup>, the largest), Chesapeake Bay and the East China Sea. Regional agreements (HELCOM for the Baltic, OSPAR for the NE Atlantic) impose nutrient load reduction targets.

**Eutrophication:** The enrichment of water bodies with dissolved nutrients (especially nitrogen and phosphorus), stimulating algal blooms whose subsequent decay depletes dissolved oxygen and produces hypoxic “dead zones” in which fish and most benthic fauna cannot survive.

## Ocean acidification

Roughly 30% of anthropogenic CO<sub>2</sub> has dissolved in surface waters, forming carbonic acid. Surface ocean pH has declined from ~8.2 in 1850 to ~8.1 today — a 0.1 unit change representing ~30% increase in hydrogen-ion concentration. By 2100 under high-emission scenarios surface pH could reach 7.7–7.8. Acidification reduces the availability of carbonate ions, making it harder for calcifying organisms (corals, shellfish, coccolithophores, pteropods) to build skeletons and shells. The Pacific oyster industry in the US Pacific Northwest was devastated by acidified upwelling in 2007–08 and has since installed pH monitoring on hatchery intakes.

**Ocean acidification:** The ongoing reduction in the pH of the ocean caused by uptake of atmospheric CO<sub>2</sub>; since the Industrial Revolution the surface ocean has become ~30% more acidic, with consequences for calcifying organisms (corals, shellfish, plankton) and for the food webs that depend on them.

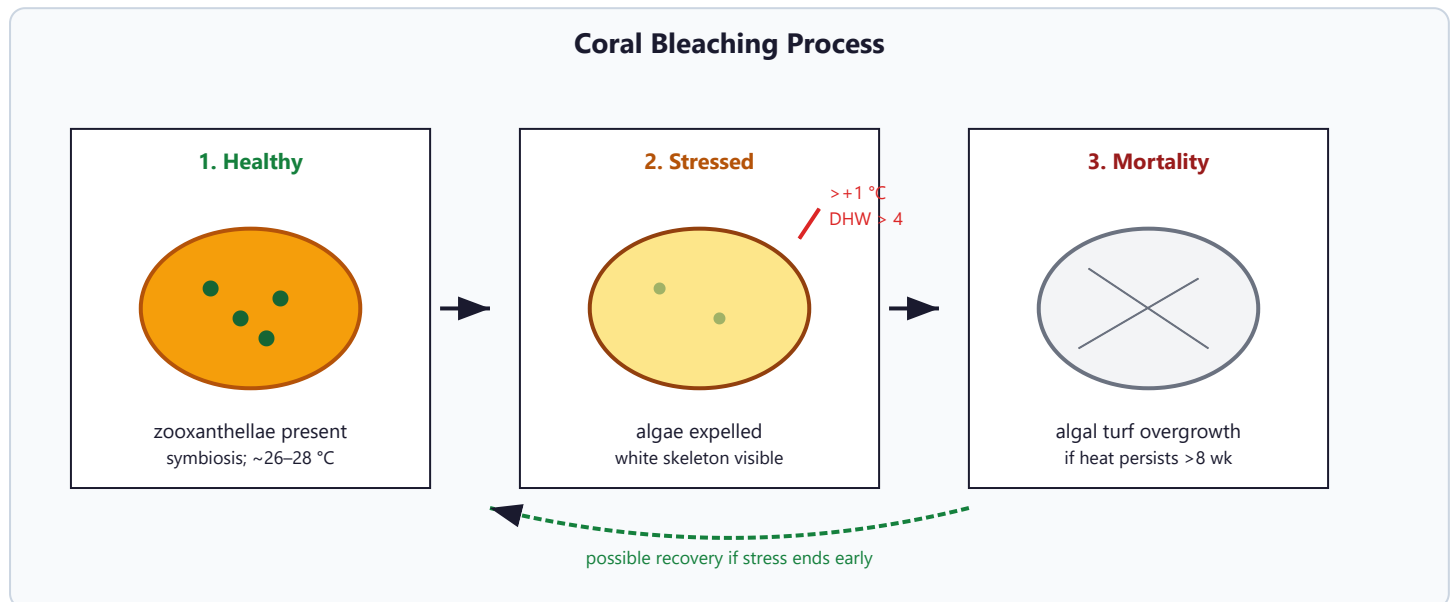
## 7. Climate Change and the Oceans

Climate change transforms every aspect of the ocean. Four linked changes deserve exam attention.

1. **Warming:** Upper-2000 m ocean heat content is rising at ~0.37 W/m<sup>2</sup>; surface temperature has warmed ~1.1 °C since pre-industrial, with >0.1 °C/decade in recent decades. 2023 and 2024 set consecutive global surface-temperature records.
2. **Marine heatwaves (MHWs):** Prolonged anomalously warm ocean conditions, defined as SST exceeding the 90th percentile baseline for >5 days. MHWs have roughly doubled in frequency since 1982. The “Blob” in the NE Pacific (2013–16) caused mass die-offs of sea stars, seabirds and whales.
3. **Deoxygenation:** Warm water holds less oxygen; global ocean O<sub>2</sub> has fallen ~2% since 1960 and oxygen-minimum zones have expanded. Fisheries in affected regions shift poleward or upward.
4. **Sea-level rise:** Currently ~3.7 mm/yr (2006–present), accelerating. Small island states (Maldives, Tuvalu, Marshall Islands) face existential threat; IPCC AR6 projects 0.44–0.76 m of rise by 2100 under an intermediate scenario (SSP2-4.5).

## Coral bleaching

Corals bleach when sustained water temperatures  $\sim 1^\circ\text{C}$  above the local summer maximum cause them to expel their zooxanthellae; without the algae they lose their colour and, if conditions persist, die. NOAA uses *degree heating weeks* (DHW) to forecast bleaching risk. Mass global bleaching events have occurred in 1998, 2010, 2014–17, and the **fourth global bleaching event declared April 2024** had affected about 84% of the world's coral-reef area, in more than 80 countries, by 2025.



### Case study: Great Barrier Reef, Australia

The Great Barrier Reef (GBR) is the world's largest coral reef system, stretching  **$\sim 2,300\text{ km}$**  along the Queensland coast and covering  $\sim 344,400\text{ km}^2$ . It contains  $\sim 3,000$  individual reefs and 900 islands and hosts  $\sim 1,500$  fish species. A UNESCO World Heritage Site since 1981; marine park managed by the Great Barrier Reef Marine Park Authority (GBRMPA) since 1975.

*Mass bleaching events:* 1998, 2002, 2016, 2017, 2020, 2022 and **2024** — seven events, with 2016, 2017 and 2024 the most severe. The 2016 event killed  $\sim 30\%$  of shallow-water corals in the northern third of the reef; the 2022 event was the first to occur in a La Niña year (previously thought protective); the 2024 event was the fifth in eight years and affected reefs across the entire length of the GBR.

*Economic significance:*  $\sim \$6.4$  billion AUD/year in tourism revenue; 64,000 jobs; cultural importance to  $\sim 70$  Traditional Owner groups whose sea country overlaps the park.

*Policy responses:* The Reef 2050 Long-Term Sustainability Plan (2015, updated 2021) coordinates

Commonwealth and Queensland action on water quality, coastal habitats, biodiversity and climate. \$3+ billion in federal funding 2015–2030. In 2021 the UNESCO World Heritage Committee considered adding the GBR to the “in danger” list; Australia’s intensive lobbying deferred the decision, the 2022 UNESCO monitoring mission then recommended an “in danger” listing, but the Committee again declined to list the reef in 2023 and 2024, asking Australia for further progress reports.

*Governance tensions:* Fundamentally the GBR’s fate is determined by global greenhouse-gas emissions, not Australian reef management. Local efforts (runoff reduction, crown-of-thorns starfish control, coral larvae seeding) are meaningful but cannot compensate for a 1.5 °C+ world. The case illustrates the limits of national conservation against a global driver.

## 8. Marine Protected Areas and Conservation

A **Marine Protected Area (MPA)** is a defined area of ocean where human activities are managed to conserve nature, cultural heritage or resources. MPAs vary enormously in protection level — from strict “no-take” marine reserves to multiple-use zones that permit fishing with restrictions. As of 2024, **~8.3% of the global ocean** is within some form of MPA, but only ~2.9% is fully or highly protected. The **30x30** target — protecting 30% of land and sea by 2030 — was adopted in the Kunming-Montreal Global Biodiversity Framework (December 2022).

**Marine Protected Area (MPA):** A clearly defined geographical space in the ocean, recognised, dedicated and managed, through legal or other effective means, to achieve long-term conservation of nature with associated ecosystem services and cultural values (IUCN definition). Protection levels range from fully protected “no-take” reserves to multiple-use zones.

| MPA                                        | Country / region    | Size (km <sup>2</sup> ) | Designated           | Protection level |
|--------------------------------------------|---------------------|-------------------------|----------------------|------------------|
| Ross Sea Region MPA                        | Antarctica (CCAMLR) | 1,550,000               | 2016 (in force 2017) | No-take (72%)    |
| Papahānaumokuākea Marine National Monument | USA (Hawaii)        | 1,510,000               | 2006, expanded 2016  | No-take          |

|                                                             |                         |                       |                                         |                                                    |
|-------------------------------------------------------------|-------------------------|-----------------------|-----------------------------------------|----------------------------------------------------|
| Marae Moana                                                 | Cook Islands            | 1,976,000             | 2017                                    | Multi-use; 50 nm no-take around islands            |
| UK Blue Belt                                                | UK Overseas Territories | >4,300,000 cumulative | 2016–present                            | Mixed; largest Pitcairn (842,000 km <sup>2</sup> ) |
| Phoenix Islands PA (now closed to commercial fishing: 2015) | Kiribati                | 408,000               | 2008; partial reopening considered 2022 | Multi-use / fluctuating                            |
| Great Barrier Reef Marine Park                              | Australia               | 344,400               | 1975                                    | ~33% no-take                                       |

Effectiveness varies widely. A 2018 global review (Sala & Giakoumi) found that fully protected marine reserves hold, on average, ~670% more fish biomass than nearby unprotected areas, but “paper parks” with no enforcement add little. The **UK Blue Belt** around British Overseas Territories (Pitcairn, Tristan da Cunha, St Helena, Ascension, South Georgia, BIOT) is one of the world’s largest MPA networks and uses satellite monitoring (OceanMind, Global Fishing Watch) to detect illegal incursions.

**Exam tip — critique MPA effectiveness:** Don’t just celebrate 8% of the ocean protected. Ask: *is it the right 8%?* Most designated area is in remote high seas where pressure is already low, while inshore biodiversity hotspots remain unprotected. Quality (enforcement, no-take area, surveillance) matters more than quantity.

## 9. The BBNJ (“High Seas”) Treaty 2023

Until 2023 the high seas — two-thirds of the ocean — had no legal mechanism to designate MPAs, no binding requirement for environmental impact assessment of high-seas activities, and no framework for sharing benefits from marine genetic resources (MGRs). After nearly two decades of negotiation (from the 2004 Working Group through the 2015 UN General Assembly resolution to five sessions of an Intergovernmental Conference 2018–23), delegates agreed the **Agreement on Marine Biological Diversity of Areas Beyond National Jurisdiction (BBNJ Agreement)** on 4 March 2023; it was adopted on **19 June 2023**, with states invited to sign from 20 September 2023. It requires 60 ratifications to enter

into force; **the 60th ratification came in September 2025**, and the Agreement entered into force on 17 January 2026.

**BBNJ Agreement (“High Seas Treaty”) 2023:** The Agreement on Marine Biological Diversity of Areas Beyond National Jurisdiction, adopted under UNCLOS in June 2023. The first legally binding instrument for biodiversity on the high seas. Covers four pillars: (i) marine genetic resources and benefit-sharing; (ii) area-based management tools including MPAs; (iii) environmental impact assessment; (iv) capacity-building and marine technology transfer.

The treaty fills a major gap: it provides a legal mechanism to designate high-seas MPAs (essential to reaching the 30x30 target), imposes EIA requirements on activities with potential significant effects in ABNJ, establishes benefit-sharing for marine genetic resources (addressing a long-standing Global South equity concern), and creates a Conference of the Parties, Scientific and Technical Body and Clearing-House Mechanism.

## 10. Deep-Sea Mining

The Area (the seabed beyond national jurisdiction) contains three prospective mineral deposits: **polymetallic nodules** (~4–15 cm concretions on abyssal plains, rich in nickel, copper, cobalt, manganese; the *Clarion-Clipperton Zone* between Hawaii and Mexico is the prime target); **polymetallic sulphides** (around active and inactive hydrothermal vents, rich in copper, zinc, gold, silver); and **cobalt-rich ferromanganese crusts** (on seamounts). The ISA has issued **~31 exploration contracts** (mostly for nodules in the CCZ) to state-sponsored companies and consortia.

The economic driver is the energy transition: nickel, cobalt, copper and manganese are all battery- and wind-turbine- metals, and projected demand exceeds terrestrial supply. The Canadian firm The Metals Company (TMC) is the most advanced private operator. In June 2021 Nauru triggered the “**two-year rule**” (Annex I, Section 1(15) of the 1994 Implementation Agreement), which obliged the ISA to finalise exploitation regulations by July 2023; that deadline was missed and ISA Council negotiations continue. In April 2025 a US executive order encouraged seabed-mining licences under US domestic law, and TMC applied to the US government — bypassing the ISA and drawing wide international criticism.

A **precautionary pause** or moratorium is backed by an increasing coalition including **France, Germany, Spain, Chile, Brazil, Canada, New Zealand, Palau, Fiji, Micronesia, Sweden, Finland and Costa Rica** (nearly 40 states by 2025). Environmental concerns include: abyssal plains are ancient, slow-recovering

ecosystems with high endemism; sediment plumes from mining can travel hundreds of km; noise and light disturb deep-sea fauna; and the climate role of seabed carbon storage is poorly understood. Major companies — European carmakers (including BMW, Volvo, Volkswagen, Renault) plus Google, Samsung SDI and Philips — have pledged not to source deep-sea minerals pending further scientific evidence.

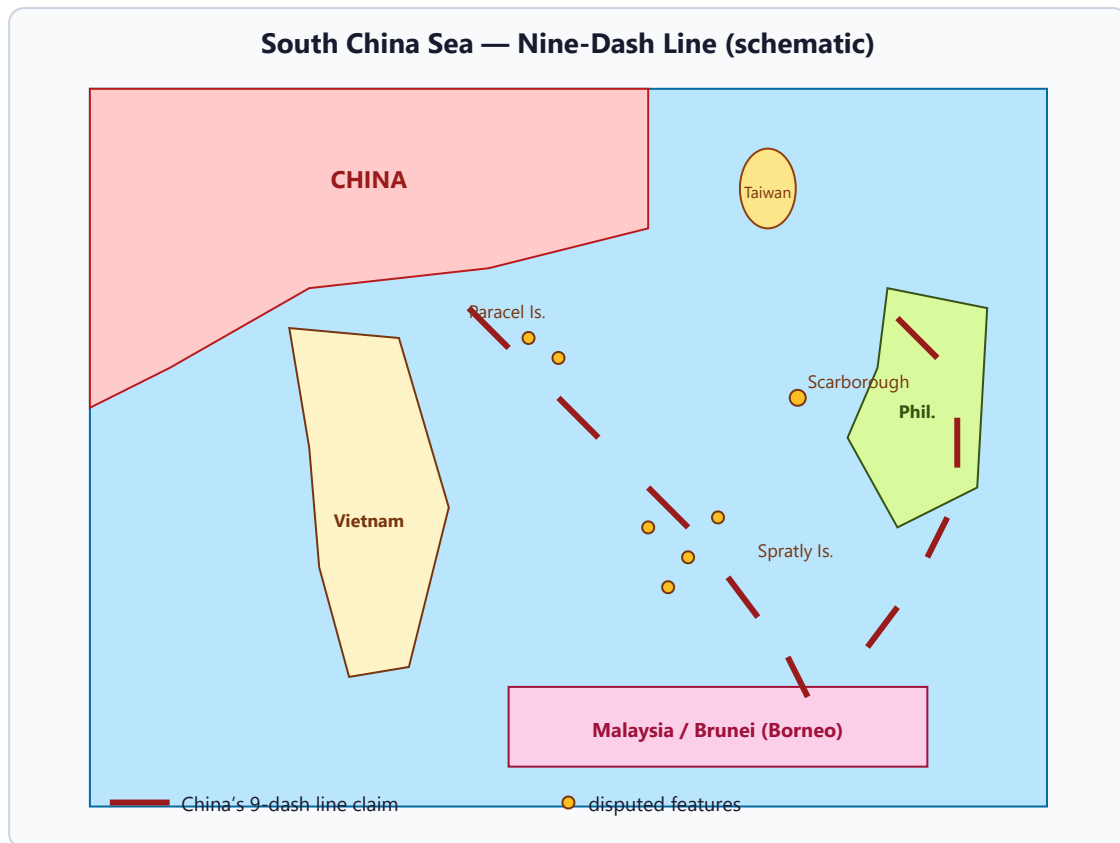
**Exam tip — evaluating governance effectiveness:** Deep-sea mining is a live example of governance *in progress*. Strong answers note (a) the ISA exists and has a clear mandate; (b) it has failed to agree exploitation rules in three decades; (c) the moratorium coalition shows states can act via soft power even without formal rules. “The institution is present but the rules are incomplete” is a neat framing.

## 11. Territorial Disputes at Sea

Where UNCLOS zones overlap, or where historic claims conflict, maritime disputes proliferate. Three are central to A-level specification: the **South China Sea**, the **Arctic Ocean**, and the **Falklands / Malvinas**.

### South China Sea

The most geopolitically significant ocean dispute. China’s claim is anchored on the **nine-dash line** (“ten-dash” in some Chinese maps), first published by the Republic of China in 1947 and inherited by the PRC, encompassing ~90% of the South China Sea. Overlapping claims involve Vietnam, the Philippines, Malaysia, Brunei, Taiwan and Indonesia. Key disputed features include the *Spratly Islands*, *Paracel Islands* and *Scarborough Shoal*.



### Case study: South China Sea territorial disputes

*Economic stakes:* ~30% of global maritime trade (~\$3.4 trillion/year) transits the South China Sea; the area holds estimated reserves of ~11 billion barrels of oil and 190 trillion cubic feet of natural gas; and it supports ~12% of the global fish catch, sustaining ~3.7 million fishing livelihoods in surrounding states.

*Chinese actions:* From 2013 China undertook massive **land reclamation** on seven reefs and features in the Spratlys, creating ~13 km<sup>2</sup> of new land (Fiery Cross, Mischief, Subi, Cuarteron, Gaven, Hughes, Johnson South). Fiery Cross now has a 3,000 m runway, deep-water harbour, radar domes and HQ-9 SAM batteries. Mischief and Subi Reefs are similarly militarised. In 2013 the Philippines filed an UNCLOS arbitration.

*2016 Permanent Court of Arbitration ruling:* On 12 July 2016 the PCA tribunal at The Hague ruled that (i) China's nine-dash line has no basis in international law; (ii) none of the Spratly features generates an EEZ of its own; (iii) China had violated Philippine sovereign rights by interfering with fishing and oil exploration. China rejected the ruling and refused to participate in the arbitration.

*Freedom of Navigation Operations (FONOPs):* The US Navy has conducted regular FONOPs since 2015 — warships sailing within 12 nm of disputed features to assert that they are not entitled to a territorial sea. ~10 operations per year 2015–2024. UK Royal Navy (HMS *Queen Elizabeth* 2021), French, Australian, Canadian and Japanese navies have followed.

*ASEAN Code of Conduct:* ASEAN and China signed a non-binding Declaration on Conduct in 2002; negotiations on a binding Code of Conduct are in their 23rd year as of 2024, with limited progress. The July 2024 ASEAN Summit pushed for completion by 2026, a target widely viewed as optimistic.

*Why it matters:* The case illustrates the limits of UNCLOS as a binding system when a major power rejects its adjudicative mechanisms, the interaction of economic, security and environmental interests, and the tension between historic-rights claims and modern treaty law.

## Arctic Ocean

The Arctic is warming at **four times the global rate**; September sea-ice extent has declined ~13% per decade since 1979. The retreat opens three governance flashpoints. First, new shipping routes: the *Northern Sea Route* (Russian Arctic) and *Northwest Passage* (Canadian Arctic) offer shorter transits between Europe and Asia; Russian NSR traffic reached ~36 million tonnes in 2023. Second, resource access: the Arctic is thought to hold ~13% of undiscovered oil and ~30% of undiscovered natural gas globally. Third, extended continental shelf claims: Russia, Canada, Denmark (via Greenland) and the US have overlapping claims centred on the *Lomonosov Ridge*, a submarine ridge that runs across the pole. Russia in 2001 (revised 2015) submitted a claim to the CLCS for a shelf extending to the Pole; Denmark and Canada have filed overlapping claims.

The **Arctic Council** (founded 1996), comprising the eight Arctic states plus six Permanent Participants representing indigenous peoples (Inuit Circumpolar Council, Saami Council etc.), is the principal regional forum — though its work was paused after Russia's February 2022 invasion of Ukraine and has resumed only at working level since 2024. A **2018 Central Arctic Ocean Fisheries Agreement** imposes a 16-year moratorium on commercial fishing in the central Arctic high seas while scientific understanding develops.

## Falklands / Malvinas

A simmering sovereignty dispute between the UK and Argentina over the Falkland Islands (Islas Malvinas). The UK has administered the archipelago since 1833; Argentina claims inheritance from Spanish colonial administration. The 1982 war (74 days, ~900 deaths) settled de facto control; in a 2013 referendum 99.8%

of Falkland Islanders voted to remain a British Overseas Territory. The dispute now centres on a 200 nm conservation and management zone around the islands, significant fishing revenues (~£40 million/year to the Falkland Islands government), and prospective oil reserves (Sea Lion field, Rockhopper plc).

## 12. Shipping and Global Trade

The ocean is the sinew of globalisation. **~80% of world merchandise trade by volume** and ~70% by value moves by sea, carried on ~50,000 commercial ships >500 GT (UNCTAD 2023). Container traffic is ~850 million TEU per year. Global shipping emits ~3% of greenhouse gases; the IMO adopted a revised strategy in 2023 targeting net-zero “by or around 2050” with 20–30% reductions by 2030 and 70–80% by 2040. A proposed IMO Net-Zero Framework (a global fuel standard with a carbon price) was approved in April 2025, but its formal adoption was postponed for a year in October 2025 after US opposition.

### Chokepoints

Roughly ~30% of world trade passes through a handful of maritime chokepoints, any of which could be closed or disrupted:

- **Strait of Hormuz** — ~20 million barrels of oil/day (~20% of global); Iran and Oman coasts.
- **Strait of Malacca** — ~94,000 ships/year; link between Indian and Pacific Oceans.
- **Suez Canal** — ~12% of world trade; Egypt.
- **Bab al-Mandab** — link between Red Sea and Indian Ocean; ~10% of world trade.
- **Panama Canal** — ~5% of world trade; affected by drought-driven freshwater shortages 2023–24.
- **Turkish Straits (Bosporus, Dardanelles)** — ~3% of world oil; regulated by Montreux 1936.
- **Danish Straits** — Baltic to North Sea, Russian oil transit.

### Contemporary security threats

- **Ever Given Suez blockage (March 2021):** The 400 m container ship ran aground in the Suez Canal on 23 March, blocking the canal for **6 days** and delaying ~\$9.6 billion of trade daily; a vivid demonstration of chokepoint fragility.
- **Red Sea Houthi attacks (2023–24):** Following the Gaza conflict, Yemeni Houthis launched missiles and drones against ships transiting the Bab al-Mandab and southern Red Sea from October 2023. Major container lines (Maersk, MSC, Hapag-Lloyd) diverted around the Cape of Good Hope, adding 10–14 days and ~\$1 million per voyage; Suez revenues fell ~50% in early 2024, and attacks continued into

2025 (two bulk carriers were sunk in July 2025). Operation Prosperity Guardian (US-led) and EUNAVFOR Aspides provide naval protection.

- **Panama drought 2023–24:** Low water levels in Gatun Lake cut daily transits from ~38 to ~22 in late 2023; climate change as an emerging shipping risk.

### Case study: Somali piracy and the international response

*Context:* Somalia has lacked a functional central government since 1991. Illegal foreign fishing and toxic waste dumping in Somali waters through the 1990s destroyed local fisheries; former fishermen turned to piracy, initially targeting illegal fishing vessels and later shifting to commercial shipping transiting the Gulf of Aden — one of the world's busiest sea lanes.

*Peak:* Attacks rose from a handful in the early 2000s to **236 attacks in 2011** (one every ~36 hours). Pirates held dozens of ships and hundreds of crew for multi-million dollar ransoms (total ransoms ~\$400 million 2005–12). At peak ~736 hostages held simultaneously.

*International response:*

- **EU NAVFOR Operation Atalanta** (December 2008–present): EU naval mission under UN Security Council Resolutions 1814, 1816, 1838, 1846. Escorts WFP food-aid convoys; protects vulnerable shipping.
- **NATO Operation Ocean Shield** (2009–2016): Complementary NATO mission.
- **Combined Task Force 151:** US-led multinational; ~30 participating nations rotating.
- **Industry self-help:** Best Management Practices (BMP-5), Privately Contracted Armed Security Personnel (PCASP), International Recommended Transit Corridor, convoy scheduling. No ship with PCASP has ever been successfully pirated.
- **Prosecution:** Seychelles, Kenya and Mauritius received pirates for trial under UNODC-supported programmes; a specialised regional prosecution centre was established in Seychelles.

*Outcome:* Attacks fell from 236 (2011) to ~0 per year by 2018–22. A modest **resurgence 2023–24** — ~8 incidents recorded in 2024 — is linked to reduced naval presence (diverted to the Red Sea) and renewed Somali economic stress.

*Governance lesson:* Piracy fell because naval, legal, industry and regional capacity responses were

combined, not because any single measure worked alone. Yet root causes — state failure, illegal fishing, unemployment — remain, and suppression is fragile.

## 12a. Institutions, Seafloor Cables, Maritime Crime and Unequal Access

**Post-1945 institutions.** The specification expects you to know the main bodies of global governance and how they touch the oceans: the *UN* (1945) and its agencies (UNESCO and its Intergovernmental Oceanographic Commission, the IMO, the FAO); the *EU* (Common Fisheries Policy, marine-waste rules); *NATO* (1949), which patrols sea lanes and since January 2025 has run the Baltic Sentry mission to protect undersea infrastructure; the *G7* (*G8* while Russia was a member, 1997–2014) and the *G20*, which set agendas on marine litter and illegal fishing; and the *G77* group of developing countries, which championed the “common heritage of mankind” principle for the deep seabed and fair sharing of marine genetic resources in the BBNJ Agreement.

**The UK as a former maritime power.** Naval supremacy underpinned the British Empire, whose trade and settlement links survive in the 56-member Commonwealth and in migration and trade ties, and in 14 British Overseas Territories that give the UK one of the world’s largest EEZs (the basis of its Blue Belt MPAs).

**Seafloor data cables.** About 1.5 million km of submarine fibre-optic cable (several hundred systems) carries ~99% of intercontinental internet and financial data. Demand from cloud computing and streaming means that technology firms (Google, Meta, Microsoft, Amazon) now fund many new cables, which converge on landing hubs and chokepoints such as the Red Sea, Egypt and the Strait of Malacca. *Risks:* most of the ~150–200 faults a year are accidental damage by fishing gear and ship anchors (in February 2024 three Red Sea cables were cut, apparently by the dragging anchor of a Houthis-damaged ship); natural hazards include submarine landslides and turbidity currents triggered by earthquakes (the 2006 Hengchun earthquake off Taiwan broke several cables) and volcanic eruptions and tsunamis (the 2022 Hunga Tonga eruption severed Tonga’s only international cable); and suspected sabotage (Baltic Sea incidents in 2023–25). Protection rests on the 1884 Convention for the Protection of Submarine Telegraph Cables, UNCLOS (Articles 113–115), cable-protection zones and industry bodies such as the International Cable Protection Committee.

**Smuggling and trafficking by sea.** Container shipping and small boats carry illicit flows — cocaine hidden in containers bound for Rotterdam and Antwerp, “narco-submarines” in the eastern Pacific, and people smuggled across the Mediterranean, the Andaman Sea and the English Channel. International

responses include the UN Palermo Protocols (2000) against migrant smuggling and human trafficking, the UNODC Global Maritime Crime Programme, EU naval operations and bilateral returns agreements.

**Unequal access to ocean resources.** 44 countries are landlocked. UNCLOS (Article 125) gives them a right of access to the sea, but they depend on neighbours' ports and transit corridors (landlocked Ethiopia, for example, relies mainly on Djibouti) and gain nothing from EEZ resources. Some coastal indigenous peoples also lose out when industrial fleets, mining or conservation zones restrict traditional fishing.

## 13. Sustainable Development Goal 14 — Life Below Water

Adopted by the UN General Assembly in September 2015 as part of the 2030 Agenda, **SDG 14** ("Conserve and sustainably use the oceans, seas and marine resources for sustainable development") has ten targets.

Progress is the weakest of any SDG goal.

| Target | Focus                                                    | Deadline | Progress (2024)                     |
|--------|----------------------------------------------------------|----------|-------------------------------------|
| 14.1   | Reduce marine pollution, esp. from land-based activities | 2025     | Worsening (plastic rising)          |
| 14.2   | Manage and protect marine and coastal ecosystems         | 2020     | Missed; ecosystems declining        |
| 14.3   | Minimise ocean acidification via enhanced cooperation    | ongoing  | Acidification accelerating          |
| 14.4   | Effectively regulate harvesting; end overfishing, IUU    | 2020     | Missed; 37.7% overfished            |
| 14.5   | Conserve at least 10% of coastal and marine areas        | 2020     | Missed (~8% by 2020; 8.3% in 2024)* |
| 14.6   | Prohibit harmful fisheries subsidies                     | 2020     | Partial — WTO Agreement 2022        |
| 14.7   | Increase economic benefits to SIDS and LDCs              | 2030     | On track for some SIDS              |
| 14.a   | Increase scientific knowledge and research capacity      | ongoing  | UN Ocean Decade 2021–30             |

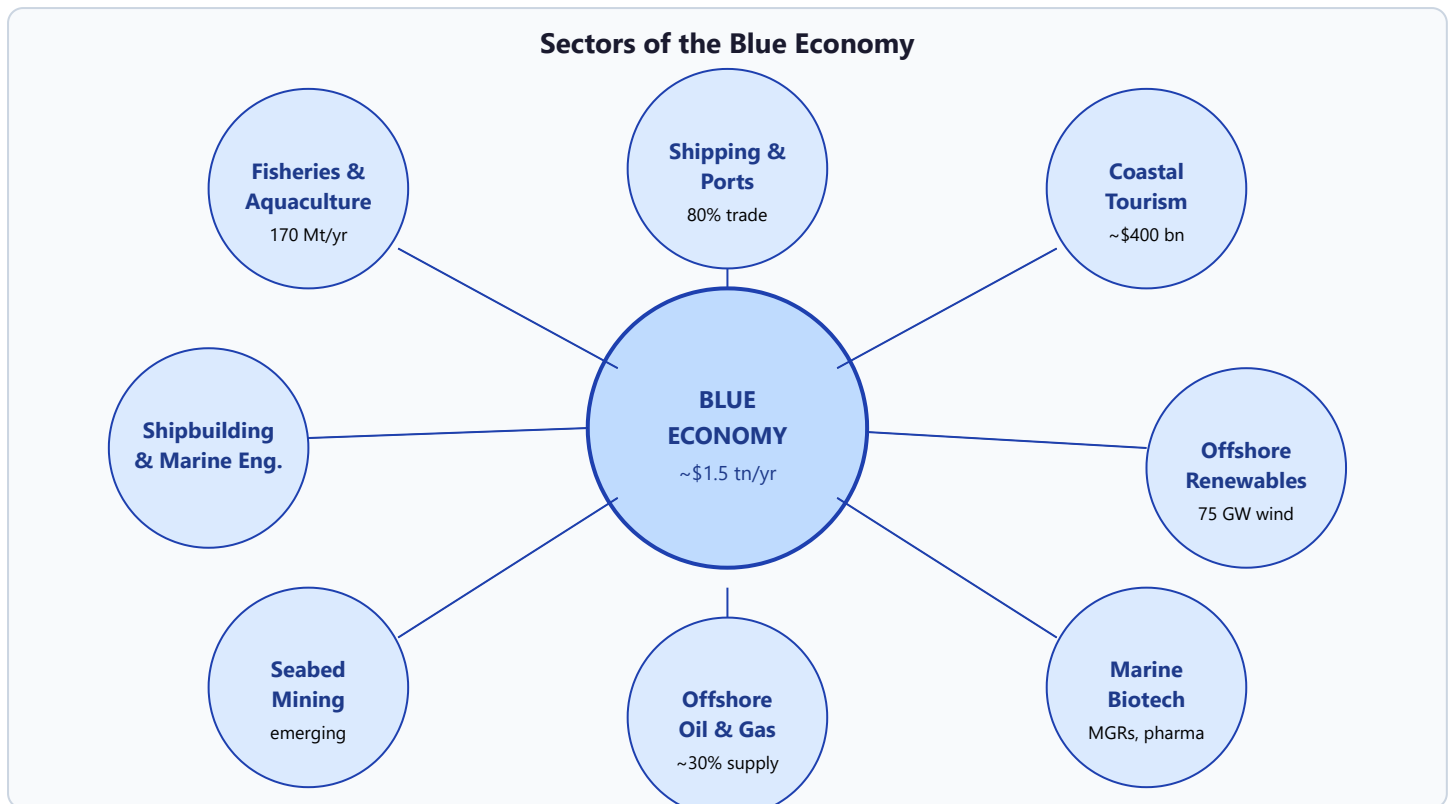
|      |                                          |         |                                        |
|------|------------------------------------------|---------|----------------------------------------|
| 14.b | Access for small-scale artisanal fishers | ongoing | Uneven                                 |
| 14.c | Implement UNCLOS                         | ongoing | Widely ratified but uneven enforcement |

\*Note: 14.5 is measured as MPA coverage, but most of this is in high seas; coastal MPA coverage (arguably more ecologically meaningful) remains below target.

## 14. The Blue Economy

The **Blue Economy** (sometimes “Ocean Economy”) is the sustainable use of ocean resources for economic growth, improved livelihoods, and ecosystem health. The World Bank estimates its global value at **~\$1.5 trillion per year** and the OECD projects ~\$3 trillion by 2030 if current growth is sustained. Seven sectors dominate.

**Blue economy:** An economic framework advocating the sustainable use of ocean resources for growth, improved livelihoods and jobs while preserving the health of ocean and coastal ecosystems. Encompasses traditional sectors (fisheries, shipping, tourism) alongside emerging ones (offshore renewables, marine biotechnology, aquaculture).



## Offshore renewable energy

Global offshore wind capacity reached **~75 GW** by end-2023 (China ~38 GW, UK ~15 GW, Germany ~8.5 GW) and is projected to exceed 250 GW by 2030. Floating offshore wind (Hywind Scotland, Kincardine) opens deeper waters. Tidal stream (MeyGen, Pentland Firth) and wave energy remain niche. Offshore wind challenges include bird and marine-mammal impacts, noise during pile-driving, and cable route conflicts with fishing.

## Marine biotechnology

Marine organisms produce unique bioactive compounds — examples include the anti-cancer drug *ziconotide* (from cone-snail venom) and heat-stable DNA-copying enzymes used in PCR testing, such as *Pfu* polymerase from the vent-dwelling microbe *Pyrococcus furiosus*. The BBNJ Agreement's benefit-sharing provisions on marine genetic resources address equity concerns: historically most MGR patents (~98%) have been filed by a handful of developed-country institutions.

## Great Pacific Garbage Patch and the Ocean Cleanup

### Case study: Great Pacific Garbage Patch and the Ocean Cleanup

*The patch:* The Great Pacific Garbage Patch (GPGP), located between Hawaii and California in the North Pacific Subtropical Gyre, is the largest accumulation of ocean plastic on Earth. A 2018 aerial and trawl survey published by The Ocean Cleanup in *Scientific Reports* estimated the patch at **~1.6 million km<sup>2</sup>** (~3× the area of France), containing **~1.8 trillion pieces** weighing ~80,000 tonnes. Composition (by mass): ~46% abandoned fishing gear (“ghost nets”), with most of the rest hard plastics, sheets and films; microplastics make up only ~8% of the mass but ~94% of the pieces. Concentrations are ~100,000× higher than the open-ocean background.

*The Ocean Cleanup project:* Founded 2013 by Dutch teenager Boyan Slat. After several system iterations, *System 002 “Jenny”* deployed July 2021 — a 800 m floating U-shaped barrier towed between two vessels, concentrating plastic at a retention zone. Upgraded to *System 03* in 2023 (2,200 m span). By late 2024 the organisation reports removing **~20 million kg (20,000 t)** of plastic cumulatively from oceans and rivers (its “Interceptor” barriers target polluted rivers, since ~1,000 rivers are estimated to carry ~80% of river-borne plastic to the sea).

*Debate:*

- **Supportive view:** Removal technology pioneers a previously absent capability; public engagement is substantial; Interceptors arguably the more impactful innovation (stopping plastic at source).
- **Critical view:** Ocean-based cleanup is energy-intensive and expensive per tonne (~\$30–100/kg) compared to prevention; critics argue that at its current scale ocean clean-up can never keep pace with new plastic input. Concerns also exist about bycatch of neuston (surface-dwelling fauna) and contribution of microplastic-fragment generation.
- **Consensus:** Upstream solutions — plastic treaty, EPR, waste-management infrastructure in Asia — must dominate. Cleanup is symbolic but cannot substitute.

*Governance link:* The patch sits in high seas — until 2023 there was no legal basis to designate MPAs there. It is also subject to no single state’s waste-management jurisdiction, making it a paradigmatic global-commons problem.

## 15. Synthesis — evaluating ocean governance

Effectiveness of global ocean governance is best assessed against four criteria. Answers that apply the framework below will earn “evaluate” marks.

| Criterion              | Strength                                                                              | Weakness                                                                                         |
|------------------------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Legal coverage         | UNCLOS near-universal; BBNJ 2023 closes high-seas gap; MARPOL, CBD, regional seas     | US non-ratification of UNCLOS; enforcement limited on high seas; patchwork of RFMOs              |
| Institutional capacity | IMO, ISA, ITLOS, FAO, UNEP; ~17 RFMOs; Arctic Council; regional bodies                | Thin budgets (e.g. the ISA secretariat); capacity disparities between states; North–South equity |
| Compliance             | Flag-state / port-state / coastal-state trio; satellite surveillance (GFW, OceanMind) | Flags of convenience; rejection of PCA rulings (China 2016); IUU persistence                     |
| Outcomes               | Tanker oil spills down; MPA coverage rising; ozone parallel success possible          | 38% stocks overfished; plastic rising; acidification accelerating; SDG 14 off-track              |

**Exam tip — use the “3L” framework:** For 20-mark essays on ocean governance, structure as **L**aw (UNCLOS, BBNJ, MARPOL), **L**imits (US non-ratification, flags of convenience, China and PCA), and **L**ocals (the role of coastal states, indigenous peoples, SIDS). Three paragraphs, each with a case, invariably outperforms an essay that merely lists institutions.

**Exam tip — contested sovereignty questions:** When asked about the South China Sea or Arctic, always distinguish (a) *legal* claims under UNCLOS (baselines, EEZ, extended shelves); (b) *historical* claims (nine-dash line, Russian Arctic); (c) *de facto* control (reclamation, ice retreat). Strong answers note that governance effectiveness depends on whether powerful states accept the legal framework.

**Exam tip — quantify wherever possible:** Numbers separate top-band from mid-band answers. Memorise one figure per topic: 71% of Earth is ocean; 12/24/200/350 nm for UNCLOS zones; 8.3% MPA coverage; 37.7% stocks overfished; 11 Mt/yr plastic input; 4.9 million barrels at Deepwater Horizon; 236 Somali attacks in 2011.

## 16. Summary Table — Five Required Case Studies

| Case study                                  | Location / date                                                 | Key facts                                                                              | Governance link                                                        |
|---------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Great Pacific Garbage Patch / Ocean Cleanup | North Pacific Gyre; 2013–                                       | 1.6 million km <sup>2</sup> ; 1.8 trillion pieces; 80,000 t; ~20,000 t removed by 2024 | High-seas pollution; no sovereign jurisdiction; future Plastics Treaty |
| Deepwater Horizon                           | Gulf of Mexico; 20 April 2010                                   | ~4.9 Mb oil; 11 dead; 87-day release; \$65 bn BP costs                                 | MARPOL; OPRC 1990; US reform of offshore safety                        |
| Great Barrier Reef bleaching                | Queensland, Australia; 1998, 2002, 2016, 2017, 2020, 2022, 2024 | 2,300 km reef; 30% coral lost 2016; UNESCO in-danger pressure; Reef 2050 Plan          | Climate → ocean; national park vs global driver                        |
| South China Sea dispute                     | SCS; 9-dash line 1947–; 2016 PCA ruling                         | ~30% world trade; 11 bn bbl oil; 7 reclaimed features; PCA ruling rejected             | UNCLOS limits with major-power non-compliance                          |
| Somali piracy                               | Gulf of Aden / Horn of Africa; peak 2011 (236 attacks)          | EU Atalanta; NATO; PCASP; ~0 by 2018; 2023–24 resurgence                               | State failure; naval + legal + industry response                       |

## 17. Key Command Words for Eduqas Component 2

Eduqas A-level command words that recur in ocean-governance questions — identify them quickly and shape your answer accordingly.

| Command | Expected response                                         |
|---------|-----------------------------------------------------------|
| Outline | Describe the main features briefly; no evaluation.        |
| Explain | Provide reasons / causes; show why something is the case. |

|                |                                                                                            |
|----------------|--------------------------------------------------------------------------------------------|
| Analyse        | Break down, identify components and relationships (often causal).                          |
| Assess         | Weigh the relative importance; give a judgment supported by evidence.                      |
| Evaluate       | Full judgement on effectiveness / success / relative importance; balanced with conclusion. |
| To what extent | Indicates a partial rather than absolute answer expected.                                  |
| Discuss        | Present different perspectives; implied weighing.                                          |

**Exam tip — final-paragraph judgment:** Ocean-governance essays almost always end with an implicit question about whether governance is working. A strong final paragraph should (a) acknowledge progress (UNCLOS, BBNJ, MARPOL), (b) identify the central weakness (enforcement on high seas, great-power compliance, climate externality), and (c) offer a nuanced “effective in X, failing in Y, untested in Z” judgement. Avoid simple “yes” or “no” conclusions.

## 18. Glossary of Key Terms

| Term                                    | Definition                                                                                                              |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Baseline                                | Low-water line along the coast, used to measure all maritime zones under UNCLOS.                                        |
| Innocent passage                        | The right of foreign vessels to transit another state’s territorial sea without hostile, fishing or polluting activity. |
| Flag state                              | The state whose flag a merchant vessel flies; responsible for enforcement on its ships.                                 |
| Port state control                      | Inspection of foreign ships in national ports to verify compliance with international conventions.                      |
| Freedom of Navigation Operation (FONOP) | A naval passage through disputed waters by a non-claimant to assert that excessive claims are not recognised.           |

|                            |                                                                                                                                                |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Common heritage of mankind | UNCLOS principle (Art. 136) that the Area and its mineral resources belong to humanity as a whole, not individual states.                      |
| Precautionary principle    | Where there are threats of serious or irreversible damage, lack of full scientific certainty shall not be used as reason to postpone measures. |
| Bycatch                    | Non-target species caught incidentally during fishing; ~40 million tonnes discarded globally each year.                                        |
| 30x30                      | Target adopted in the 2022 Kunming-Montreal Global Biodiversity Framework to protect 30% of land and sea by 2030.                              |
| Degree heating weeks (DHW) | Cumulative measure of coral thermal stress; >4 DHW triggers significant bleaching, >8 DHW mass mortality.                                      |
| Ghost net                  | Abandoned, lost or discarded fishing gear that continues to catch marine life; ~10% of ocean plastic.                                          |
| SIDS                       | Small Island Developing States — ~39 states particularly dependent on ocean resources and vulnerable to sea-level rise.                        |

The ocean is simultaneously humanity's largest ecosystem, its busiest transport medium, its biggest climate regulator and its most poorly governed global commons. Twentieth-century law (UNCLOS, MARPOL, CBD) built a near-universal legal framework; the twenty-first century's challenge is to close remaining gaps (BBNJ done, plastics treaty pending, deep-sea mining rules overdue) and — above all — to integrate ocean governance with climate governance. A Great Barrier Reef that bleaches every second year, an AMOC that may weaken past tipping, and small-island states facing submergence are reminders that the ocean's physical system sets the outer boundary of what governance can achieve.

## 19. Practice Essay Plans

The following outlines are provided as templates for the most common question-types. Full essays would substitute case-specific detail; the structures below show how to deploy the material in this topic guide.

**“Evaluate the effectiveness of UNCLOS as a framework for global ocean governance.” (20 marks)**

1. **Introduction:** Define UNCLOS (169 parties, 1994 entry into force); preview judgment — comprehensive on *zones* but uneven on *enforcement*.
2. **Strengths:** Near-universal ratification; legal clarity on zones; institutional architecture (ISA, ITLOS, CLCS); dispute settlement (ITLOS Bangladesh/Myanmar 2012); adaptable through implementing agreements (BBNJ 2023).
3. **Limits:** US non-ratification; rejection by major powers (China & PCA 2016); flags-of-convenience weaken flag-state enforcement; no role in climate externalities; high-seas IUU fishing persists.
4. **Evaluation:** Strongest on zonal delimitation and shipping; weakest on high-seas enforcement and compliance by non-accepting powers.
5. **Conclusion:** UNCLOS is a necessary but not sufficient framework — its success rises or falls on implementing agreements (Plastics, BBNJ) and political will.

## “To what extent are Marine Protected Areas an effective response to ocean biodiversity loss?” (20 marks)

1. **Introduction:** Define MPA (IUCN); quote 8.3% coverage (2024); preview: effective where strictly enforced, limited where “paper parks” or facing global drivers.
2. **Effective cases:** Papahānaumokuākea (no-take, satellite monitoring); Ross Sea (1,550,000 km<sup>2</sup>); UK Blue Belt (Pitcairn); 2018 global review — ~670% more fish biomass in fully protected reserves.
3. **Limited cases:** Great Barrier Reef Park cannot defend against warming; Phoenix Islands reopened 2022 under fiscal pressure; most “MPA” area is low-pressure remote high-seas.
4. **30x30 target & BBNJ 2023:** Institutional progress but ratification and enforcement lag.
5. **Conclusion:** Effective against *local* biodiversity pressures (fishing, extraction) but powerless against *global* drivers (warming, acidification) — MPAs must be paired with climate action.

## “Assess the geopolitical significance of the South China Sea disputes.” (20 marks)

1. **Economic stakes:** ~30% world trade; 11 bn bbl oil; 12% global fish catch.
2. **Competing claims:** Nine-dash line (1947); Philippines, Vietnam, Malaysia, Brunei, Taiwan overlapping EEZs.
3. **Militarisation:** Reclaimed islands (Fiery Cross, Mischief, Subi); HQ-9 SAMs; 3,000 m runways.
4. **Legal frame:** 2016 PCA ruling; Chinese rejection; US FONOPs; UK, French, Japanese support.

5. **Regional dimension:** ASEAN Code of Conduct in 23rd year of negotiation; Vietnam-Philippines 2024 cooperation agreement.
6. **Conclusion:** Significance both *local* (ASEAN cohesion, fisheries) and *systemic* (test of rules-based order; US-China strategic competition).

## 20. Quick-Reference Facts Sheet

| Domain                          | Key number                                        | Year         |
|---------------------------------|---------------------------------------------------|--------------|
| Ocean surface area              | 361 million km <sup>2</sup> (71% of Earth)        | —            |
| Ocean volume                    | 1.335 billion km <sup>3</sup>                     | —            |
| Average depth                   | 3,682 m                                           | —            |
| Mariana Trench (deepest)        | 10,935 m (Challenger Deep)                        | —            |
| UNCLOS parties                  | 169                                               | 2024         |
| BBNJ signatures / ratifications | 60th ratification Sept 2025; in force 17 Jan 2026 | 2026         |
| Fish stocks overfished          | 37.7%                                             | FAO 2024     |
| Capture + aquaculture output    | ~170 million tonnes/yr                            | FAO 2024     |
| Plastic entering ocean          | ~11 million tonnes/yr                             | 2024         |
| MPA coverage                    | ~8.3% global ocean (~2.9% fully)                  | 2024         |
| Ocean warming (surface)         | ~0.11 °C/decade since 1970                        | IPCC AR6     |
| Ocean pH change                 | −0.1 units since 1850 (~30% more acidic)          | —            |
| Sea-level rise                  | ~3.7 mm/yr                                        | 2006–present |
| ISA exploration contracts       | ~31                                               | 2024         |

|                               |                         |             |
|-------------------------------|-------------------------|-------------|
| Global shipping GHG share     | ~3%                     | 2023        |
| Seaborne share of world trade | ~80% volume; ~70% value | UNCTAD 2023 |

Use the facts sheet to rehearse the headline numbers you can drop into opening paragraphs — the surest signal of A-grade command of the topic.

## 21. Cross-topic Links to Other Eduqas Themes

Ocean governance is exceptionally cross-cutting. Strong Eduqas candidates connect it explicitly to the other themes studied under Component 2 and the physical themes of Component 1.

- **Topic 1.1 — Coastal landscapes:** Sea-level rise and ocean warming feed back into coastal erosion, salt-marsh retreat, and the viability of soft-engineering schemes; MPAs and integrated coastal zone management (ICZM) are applied governance instruments at the coast.
- **Topic 2.1 — Water cycle and water insecurity:** The ocean holds 97% of global water and is the engine of the hydrological cycle; changing evaporation, salinity and AMOC strength alter precipitation regimes far inland. Maritime shipping is now also a water-quality stressor in port cities.
- **Topic 2.2 — Carbon cycle and energy security:** The ocean is the largest active carbon reservoir (~38,000 Gt), absorbing ~30% of anthropogenic CO<sub>2</sub>; offshore wind is now a mainstream energy-security answer in the UK (~15 GW capacity, >30% of electricity on windy days) and the North Sea has transitioned from hydrocarbon province to renewable showcase.
- **Topic 2.3 — Global migration:** Maritime routes are central to irregular migration — the central and western Mediterranean corridors, the English Channel, the Andaman Sea — and responsibility is contested between coastal states, flag states and rescue NGOs operating under the SAR Convention 1979 and SOLAS obligations.
- **Component 3 — Contemporary Themes:** Ocean governance is a live case study for evaluating whether the post-1945 rules-based order can cope with 21st-century challenges (China, climate, commons).

**Exam tip — synoptic answer-building:** Eduqas increasingly rewards synoptic links. A paragraph that joins ocean acidification (topic 2.4) to the global carbon cycle (2.2) to reef bleaching impacts on

coastal economies and migration (1.1 / 2.3) earns more than three separate paragraphs on each theme. Practise joining two or three topics into a single causal chain.

## 22. Further Reading / Recommended Sources

For wider reading beyond the textbook — examiners notice when candidates cite contemporary reports or named scholars.

- UN FAO — *The State of World Fisheries and Aquaculture (SOFIA) 2024*. Primary source for fisheries statistics.
- UNEP — *From Pollution to Solution: A global assessment of marine litter and plastic pollution (2021)*.
- IPCC — *Special Report on the Ocean and Cryosphere in a Changing Climate (SROCC, 2019)* and AR6 WGI (2021).
- IOC-UNESCO — *Global Ocean Science Report 2020*.
- World Bank — *The Changing Wealth of Nations 2021 — Blue natural capital*.
- Ocean Panel — *Transformations for a Sustainable Ocean Economy (2020)*.
- Authors to cite: Callum Roberts (*The Unnatural History of the Sea*, 2007), Sylvia Earle (*Mission Blue, The World Is Blue*, 2010), Daniel Pauly ("Fishing down marine food webs", *Science* 1998).
- Live data dashboards: Global Fishing Watch; Marine Traffic; NOAA Coral Reef Watch; Ocean Health Index.

Ocean governance is, finally, a moving target. Between the 2023 BBNJ adoption, the stalled Plastics Treaty, the unresolved ISA Mining Code and the 2024 global coral-bleaching event, the next three years will reshape what we have studied. Keep current — a single up-to-date reference ("after the failed August 2025 Geneva plastics-treaty talks...") signals serious engagement and lifts answers into the top band.